NUS IT5006 · TEAM 10 · PHASE 3 FINAL PRESENTATION

Catch the bad review before it's written.

How **100,000 Brazilian e-commerce orders** revealed why unhappy buyers slip past support — and **three simple interventions** that reach them in time.

FOUR MOMENTS IN EVERY ORDER'S LIFE

1. **Checkout** — Quote honest date
2. **Promised date** — Speak first if late
3. **Delivery** — Call top-10 risk list
4. **Review** — 1–2★ or 5★ posted

Pan Pinyou · **Li Yudan** · **Zhu Gengyu**

Olist Brazilian E-Commerce · 99,441 orders (2016–2018) · AY2026/27 Sem 1



LIVE INTERACTIVE APP

team10-it5006-olist.streamlit.app

Scan or click to try Predict & Replay live ↗

This review arrived before the parcel did

Order #e1320973 · Health & beauty item (R$65 + R$28 freight) · 2,267 km from São Luís (MA) to São Gonçalo (RJ)



2,267 km cross-country haul · Promised ~21 days, took 31.8 days

16 Jul 2018 — Day 0

Order placed at checkout

Olist promises delivery by **7 August (~21-day window)**.

7 Aug 2018 — Day 22

Promised date passes — no parcel

Still in carrier transit. **Zero proactive warning** sent to the buyer.

9 Aug 2018 — Day 24

Olist's automated survey fires anyway

Triggered by the **promised calendar date** — not by actual delivery.

11 Aug 2018 — Day 26

Customer submits a 1-star review

★☆☆☆☆

Frustrated and still empty-handed, the buyer locks in **1★ while waiting**.

17 Aug 2018 — Day 32

Parcel finally arrives (10.5 days late)

Too late for recovery — the damage on the product page is **already done**.

When the box reached the door, the 1-star review was already **5.8 days old.**

Any apology triggered at delivery was dead on arrival. Was this a rare fluke — or a marketplace-wide flaw?

A broken promise turns into a bad review

Across the entire marketplace, crossing the promised delivery date multiplies 1–2★ reviews nearly sixfold.

DELIVERED ON TIME

9.2%

leave a 1–2★ review

87,902 orders (92.0%)
Baseline product & fit issues

5.9×

DELIVERED LATE

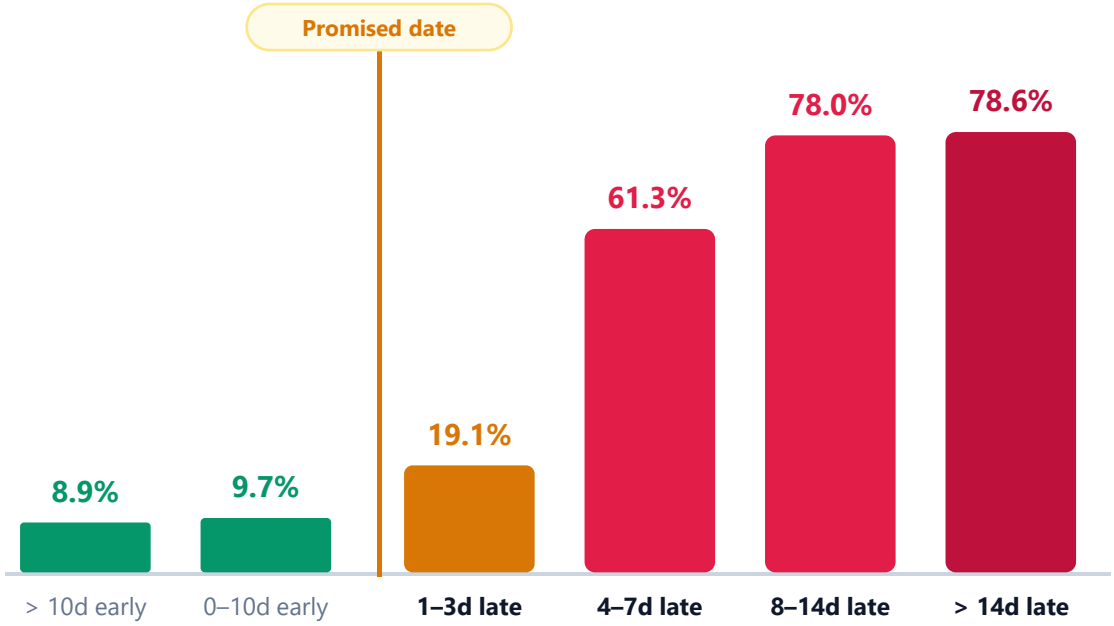
54.0%

leave a 1–2★ review

7,659 orders (8.0%)
More than half turn unhappy

How fast anger climbs once the promised date is crossed

Share of orders leaving 1–2★, grouped by arrival vs. promised date



Sharp cliff at Day +4: 1–3 days late doubles bad reviews (19.1%); 4+ days late jumps past **61%–79%**.

WHY THIS HURTS THE WHOLE MARKETPLACE

Only **1 in 12 parcels (8.0%)** arrives late — yet they generate **34% (one-third)** of every 1–2★ review on Olist.

- **Lasting revenue drag:** On Olist's own data, bad logistics reviews directly depress a seller's future sales & ratings (Deshpande & Pendem, 2023).

For a late parcel, the review comes first

Comparing review timestamps against delivery scans reveals two completely different intervention windows.

ON-TIME PARCELS (11 IN 12)

66% of all 1–2★ reviews

Wrong, damaged, or disappointing items — **99.5% review AFTER delivery.**

≈ 2.1 DAYS TO CALL
Recovery window open!



MEDIAN WAIT

+2.1 days

after delivery scan
before buyer writes

LATE PARCELS (1 IN 12)

34% of all 1–2★ reviews

Survey fires at the promised date — **buyer rates while still waiting!**

SURVEY FIRES EARLY
1★ written 4.5d early!



ALREADY WRITTEN

87.3%

of late 1–2★ reviews
submitted before arrival

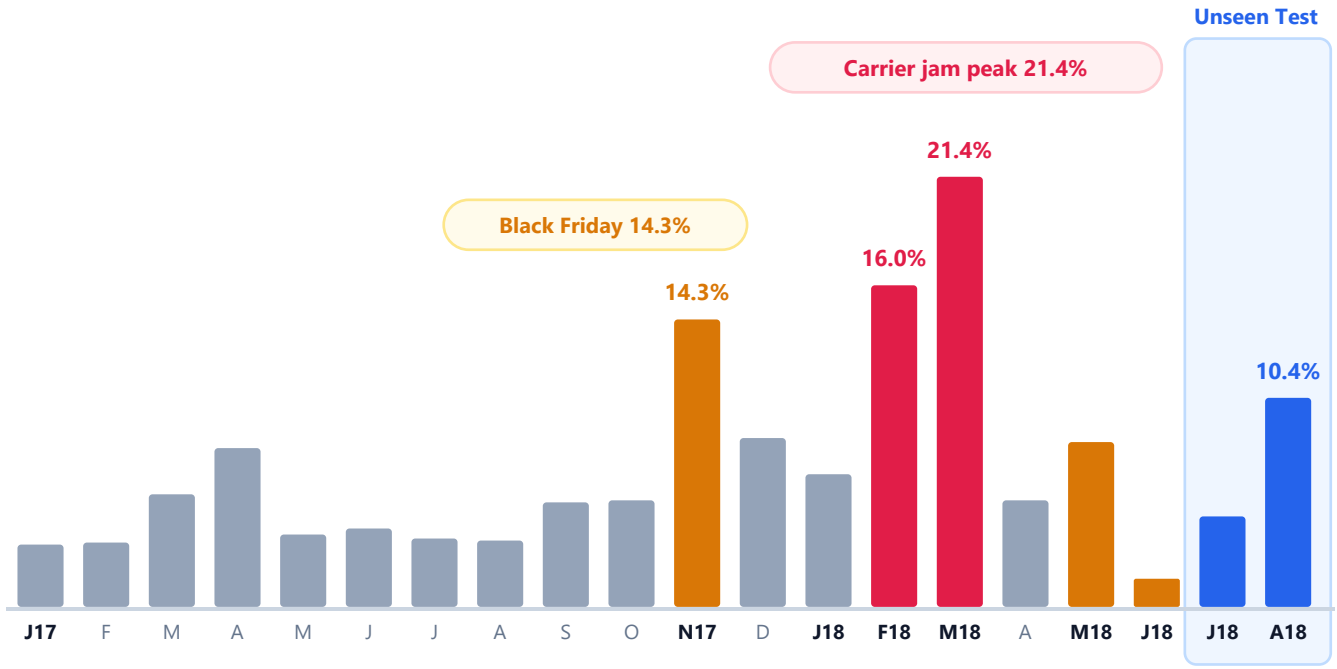
THE REFRAME: Waiting until delivery misses **87.3% of late-order bad reviews**. Olist needs **earlier interventions for late parcels** AND **fast triage at delivery for on-time ones.**

Why promises break: a moving, shock-hit network

Brazil's delivery network sped up 2× while suffering sudden carrier jams & strikes — breaking static rules and leaky tests.

Monthly late-delivery rate across 20 months (Jan 2017 – Aug 2018)

Sudden logistics shocks hit right when Olist tightened promised windows



May 2018 truckers' strike: Olist temporarily doubled promised windows (**19.5 → 36.8 days**), hiding June lateness (1.4%) at the cost of massive checkout padding.

1 · NETWORK SPEED DOUBLED

16.9 d → 7.7 d average delivery

Transit sped up 2× by Aug 2018, yet **75% of total time lives with the carrier** after seller handoff — old averages go stale within weeks.

2 · PADDED PROMISES BACKFIRE

~11 extra buffer days on average

To stay safe, Olist quoted huge buffers (9.8d error) — yet still missed **1 in 5 orders (21.4%)** when regional carrier slowdowns struck.

3 · NO PEEKING AT THE FUTURE

Fake 1.00 AUC vs. honest future test

Including post-event timestamps inflates validation AUC from **0.59 to a fake 1.00**. Every result we show is tested strictly on unseen Jul–Aug 2018 orders.

Three moments before the review — three simple tools

We didn't win by stacking fancier algorithms. We matched a simple, honest tool to each moment Olist can still act.

1 · AT CHECKOUT
Operations Team

Delivery-time forecast & derived delay risk

Predicts how many days this route & parcel will really take using only checkout info.

SOLVES:
Replaces padded 24-day guesses with honest dates & flags risky routes early.

2 · ON PROMISED DATE
Customer Care Team

Just the calendar — no ML model needed

Triggers automatically the instant a promised date passes with no delivery scan.

SOLVES:
Apologises first & pauses the review email before 87.3% of late 1–2★ are written.

3 · AT DELIVERY
Customer Experience Team

Daily review-risk call-list ranker

Scores the day's orders so agents call the ~10 buyers most likely to rate 1–2★.

SOLVES:
Catches damaged, wrong, or delayed orders inside the ~2.1-day post-delivery window.

AT REVIEW
Too Late

12.8%

(~1 in 8 delivered orders)

end in a permanent 1–2★ rating once the customer clicks submit.

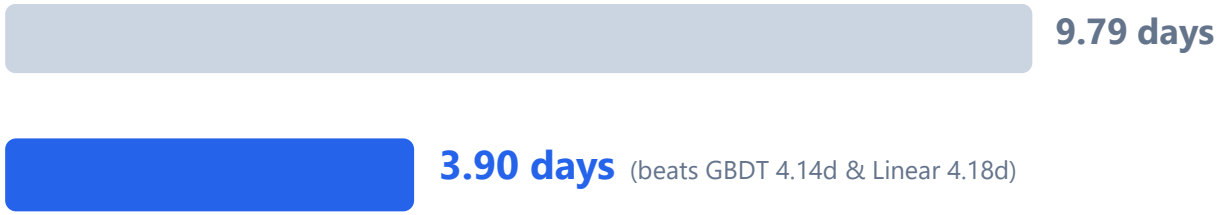
14.6% across all reviewed orders including undelivered.

WHY FANCY MODELS WEREN'T THE POINT: Even a single decision tree captures **96% of the delivery-triage signal**, and our middle intervention uses **zero ML**. Everything ships in one live **scikit-learn + Streamlit** app.

Honest dates at checkout — and why rushing loses money

Tested on 12,630 unseen future deliveries: our forecast cuts date error by 60% and tells Olist what NOT to spend money on.

AVERAGE DELIVERY-DATE ERROR AT CHECKOUT (MAE) · Lower is closer to actual arrival



60%

closer dates

Costs R$0 to quote honestly at checkout

✓ WHAT WORKS AT CHECKOUT: FREE EARLY WARNING

- **Forecasting days beats yes/no classifiers:** Deriving delay risk from our day forecast lifts PR-AUC **2.9× vs 1.2×** (ROC-AUC 0.782 vs 0.584).
- **Top 10% catches 35% of late orders (3.5×):** Our opening São Gonçalo order ranked **#23 of 238 delivered orders** on 16 July.
- **Heads-up banner above ≥ 11% risk:** Pre-warns **83% (5 in 6)** future late parcels at near-zero messaging cost.

✕ HONEST FINDING: DO NOT PAY R$12 FOR EXPRESS SHIPPING

27–28% precision < 30% break-even

At **R$12 express upgrade** vs. **R$40 bad-review cost**, at least **30% (12 / 40)** of rushed parcels must truly be late just to break even.
Why? 75% of transit happens after checkout inside the carrier network. **Save the R$12 until a live mid-transit scan stalls.**

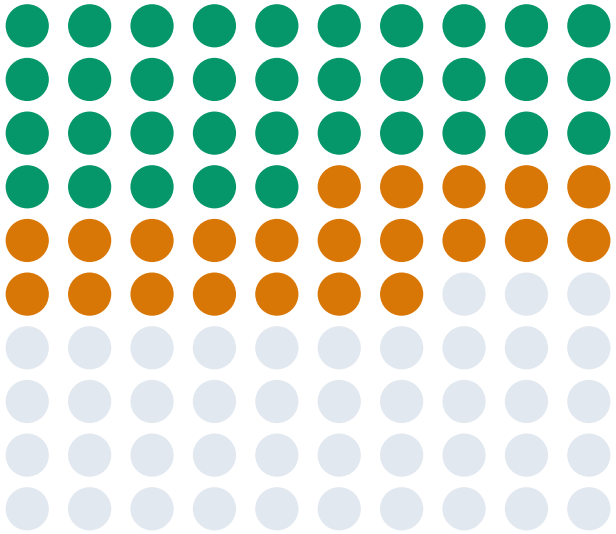
Ten calls a day find 5× more unhappy customers

Across 62 unseen test days (12,855 reviewed orders), calling just 10 ranked buyers/day turns 11% random hits into 57%.

OUR DAILY TOP-10 CALL LIST

57 / 100 calls

reach a real 1–2★ customer (57.2%)

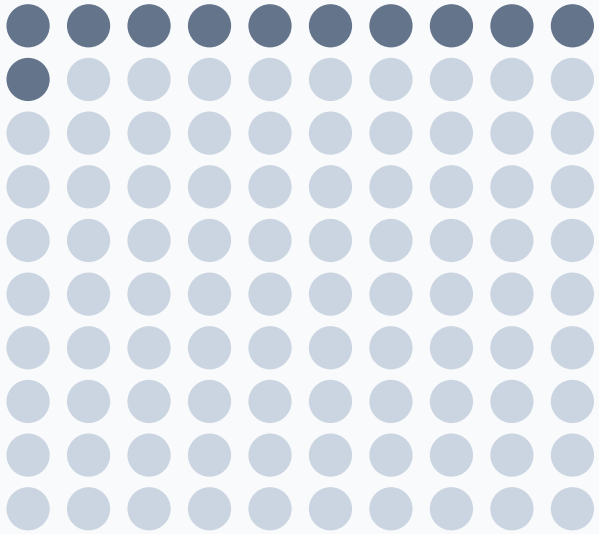


- **35** not written before arrival
- **22** wrote waiting
- **43** satisfied (3–5★) · 614 total calls over 62 days

CALLING AT RANDOM

11 / 100 calls

reach a 1–2★ customer (11.0% base)



89 of 100 calls wasted on happy buyers.

Below break-even (loses money).

PRECISION LIFT AT SMALL BUDGETS

5.2× at 10 calls/day (68% at top 5%)

Powered by delay vs. promise, multi-unit baskets, high freight share & category history.

WHAT ABOUT DELIVERED PARCELS ONLY?

45% precision (delivered only)

Without the never-delivered orders the daily top-10 still hits **45.1% unhappy buyers** (vs 9.6% at random); strictly on-time parcels: top 5% = 28% (3.5×), where **99.5% wait ~2.1 days after delivery!**

LOW CALL BREAK-EVEN (R$5 CALL : R$40 REVIEW)

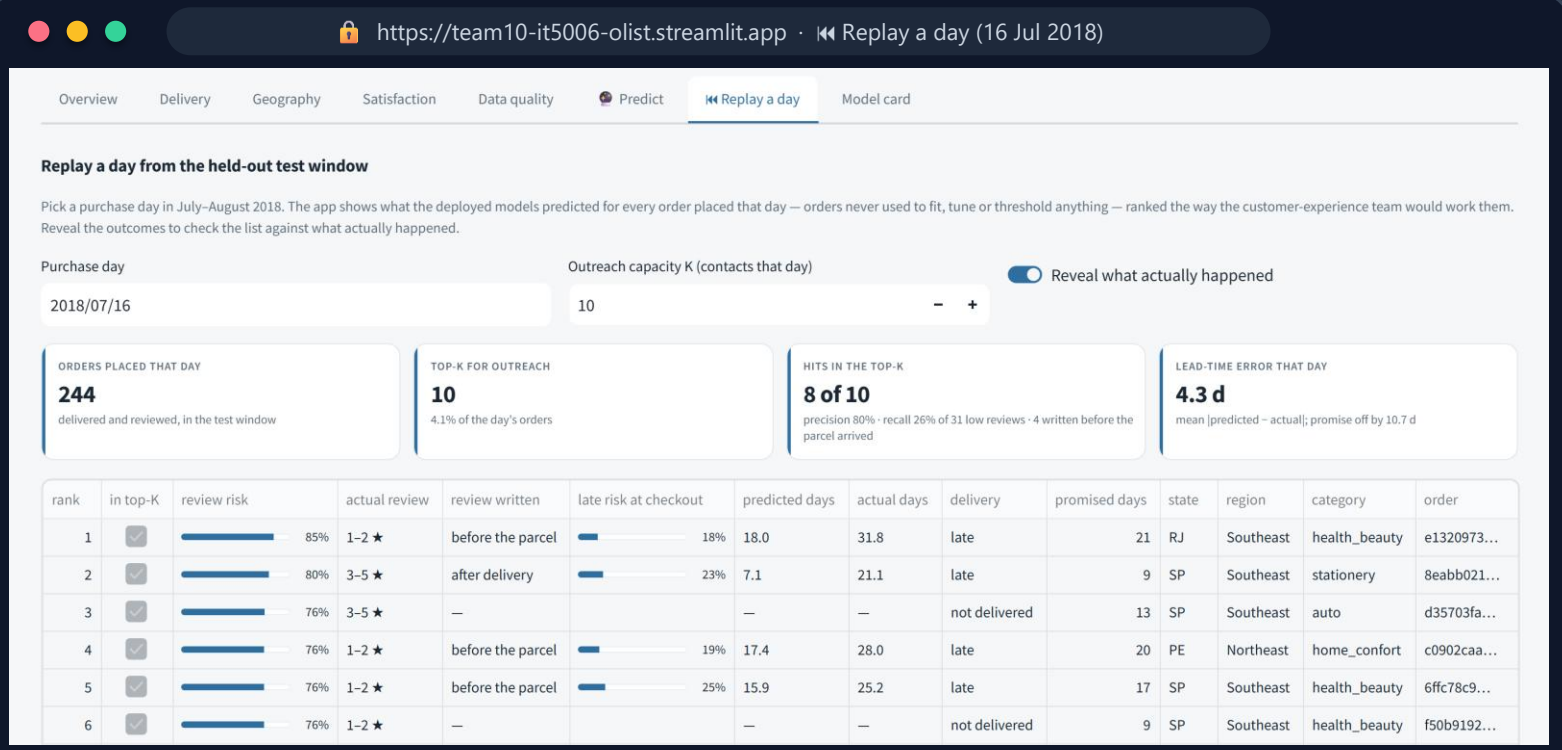
Only 13 hits / 100 needed

57 total hits (and 35 non-pre-empted) clear the 12.5% bar with wide safety margin.

▶ Open Live App ↗

Replay 16 July 2018 blind — then reveal reality

Our deployed 8-tab Streamlit app lets Operations & CX score single orders, upload CSV batches, or replay any of 62 test days.



RANKED #1 ON THAT MORNING'S LIST (85% RISK): **São Gonçalo (RJ) · health_beauty**

Our opening Slide 2 customer (#e1320973)! Flagged #23 of 238 delivered at checkout, and #1 of 244 reviewed on the call list.

16 JULY 2018 (244 REVIEWED ORDERS)

8 of 10 real 1–2★

caught in top-10 calls (vs ~1 by random)
Checkout error: 4.3 d vs Olist's 10.7 d

WHY THOSE 8 WERE UNHAPPY

- **4 buyers** wrote 1★ while still waiting for a late parcel (incl. #1 São Gonçalo).
- **4 buyers** never received their parcel at all (shipped, invoiced or cancelled).

ACROSS ALL 62 UNSEEN TEST DAYS

57 / 100 average precision

Shows why Promised-Date care + Delivery calls work together!

Four changes Olist can launch on Monday

Three proactive actions across the order timeline — plus one costly reflex to avoid.

1 · AT CHECKOUT (OPERATIONS)

Quote honest dates & warn ≥ 11% risk

Replace padded promises with our checkout forecast (**60% smaller error: 9.79 → 3.90 d**) and show a free heads-up banner above 11% risk — pre-warning **83% (5 in 6)** future late orders.

Cost: ~R$0 · Immediate trust gain

2 · ON PROMISED DATE (CUSTOMER CARE)

Speak first & pause the review email

No ML needed: the day a promised date passes with no delivery scan, **send a proactive apology + status update** and hold the review survey until actual arrival — closing the window behind **87.3% of late 1–2★**.

Cost: Automated trigger · Stops pre-arrival 1★

3 · AT DELIVERY (CUSTOMER EXPERIENCE)

Call the daily top-10 risk list

Hand support the top ~10 orders each morning (**57% precision = 5.2× random**; **45% on delivered parcels alone**) while buyers still wait ~2.1 days before reviewing.

Cost: ~R$5/call · Targets multi-item & delayed orders

4 · AT CHECKOUT (DISCIPLINED GUARDRAIL)

Do NOT pay R$12 to rush on checkout flag

Checkout-only precision plateaus at **27–28% across top-K tiers** — below the **30% break-even (R$12 / R$40)**. Save express upgrades until a live mid-transit carrier scan stalls.

Saves budget · Prevents negative-ROI upgrades

KEEPING IT FRESH IN PRODUCTION: **Refit monthly on a rolling 12-month window** (because transit speed halved in 6 months) and **monitor weekly late-rate spikes** during carrier strikes or Black Friday.

Worth R$775–R$1,466 / 1k orders — test before scaling

Historical data proves we find the right unhappy customers; a 6-week A/B pilot proves how many calls change a mind.

NET VALUE CEILING (R$5 CALL VS R$40 BAD REVIEW)

R$775 – R$1,466 / 1k orders

- **R$775** conservative ceiling on the **35/100 share** not confirmed written before arrival (R$433 at fixed 10 calls/day).
- **R$1,466** full ceiling across all hits once **Promised-Date survey pause** keeps late orders reachable.

THE HONEST CATCH IN HISTORICAL DATA

Nobody in the 2016–2018 log was ever called.

Offline test sets measure **who ends up unhappy (57.2%)** — not **how often a real phone call changes a 1★ into 4–5★**. Scaling blind would be irresponsible.

THE 6-WEEK RANDOMIZED EXPERIMENT (ZERO NEW SOFTWARE)

Split each morning's top-10 call list 50 / 50 for 6 weeks (191 orders per arm)

5 RANDOMLY CALLED / DAY
Agent checks in & resolves issue



vs

5 HELD AS CONTROL / DAY
Business-as-usual (no call)

TODAY (NO CALL)
57.2%
leave 1–2★ in control

BREAK-EVEN (~1 IN 5)
≤ 44.7%
covers R$5 call cost

6-WK TARGET (1 IN 4)
42.9%
80% power · +R$0.72/call

PLUS ARM B (PROMISED-DATE OVERDUE ORDERS): Randomize **proactive apology + holding the survey until arrival** on overdue parcels to measure pre-arrival 1★ reduction directly.

Three lessons we would tell the next analytics team

What mattered most across three phases of cleaning, modelling, and deploying on 100,000 real orders.

01 TIMING BEATS FANCY MODELS

Check when the outcome happens first

87.3% of late-order bad reviews were written **before the parcel even arrived**. Matching simple tools to the right moment — checkout, promised date, and delivery — mattered infinitely more than squeezing 0.001 AUC from a complex ensemble.

02 JUDGE STRICTLY ON THE FUTURE

Real networks drift and get hit by strikes

Delivery time halved while carrier jams and a national strike whipsawed lateness. Peeking at future timestamps gives a **fake 1.00 AUC (vs 0.59 honest)**. Always test forward in time and refit monthly.

03 DESIGN FOR BLIND SPOTS

Say clearly what data cannot prove yet

Checkout flags plateau at **27–28% (< 30% rush break-even)**, and offline logs can't prove human call persuasion. Good analytics tells leadership both where NOT to spend R$12 and how to run the 6-week pilot.

RESPONSIBLE DEPLOYMENT GUARDRAIL: **Use risk scores ONLY for proactive customer care & honest date quotes — NEVER to penalise or fine sellers.**
75% of delivery time happens inside third-party carriers after seller handoff, and Northeast / North routes face structural distance disadvantages.

THANK YOU · 2-MIN LIVE DEMO & Q&A

The review arrived before the parcel.

Olist can get there first.



1. Checkout
60% closer date

2. Promised date
Speak first + hold survey

3. Delivery
Top-10 calls (5.2×)

4. Review
Saved before 1★!

▶ LIVE STREAMLIT APP

team10-it5006-olist.streamlit.app

Click to launch 2-min live demo ↗

GITHUB REPOSITORY

Full reproducible code, tests & Phase 3 report ↗

NEXT UP · 2-MINUTE LIVE DASHBOARD DEMO · Click any step to launch live app ↗

01 · REPLAY A DAY
16 Jul 2018 blind → Reveal
Catch 8 of 10 real 1–2★ (#1 = São Gonçalo!)

02 · PREDICT LIVE
Score MA → RJ at checkout
Honest date vs promise + ≥11% warning

03 · MODEL CARD
What-if cost & pilot planner
Test R$5 call vs R$40 review & 6-wk RCT

↺ Slide 2 · Order #e1320973 ↺ Slide 4 · Timing Twist ↺ Slide 8 · 5.2× & On-Time ↺ Slide 11 · 6-Wk A/B Pilot

Team 10 · Pan Pinyou & Li Yudan & Zhu Gengyu | NUS IT5006 Fundamentals of Data Analytics | Now switching to our 2-minute live Streamlit demo →